

4.7.51

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Question

Find the equation of the plane passing through $(-1, 3, 2)$ and perpendicular to the planes $x + 2y + 3z = 5$ and $3x + 3y + z = 0$.

Solution

Normals of the given planes are

$$\vec{n}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad (1)$$

$$\vec{n}_2 = \begin{pmatrix} 3 \\ 3 \\ 1 \end{pmatrix} \quad (2)$$

$$\vec{n}_1 \cdot \vec{n}_2 = 1 \cdot 3 + 2 \cdot 3 + 3 \cdot 1 = 16 \quad (3)$$

$$|\vec{n}_1| = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14} \quad (4)$$

$$|\vec{n}_2| = \sqrt{3^2 + 3^2 + 1^2} = \sqrt{19} \quad (5)$$

Let the required plane have normal vector $\vec{n}$

Since it is perpendicular to both given planes:

$$\vec{n}_1 \cdot \vec{n} = 0, \quad \vec{n}_2 \cdot \vec{n} = 0. \quad (6)$$

That is,

$$\vec{n}_1 \cdot \vec{n}_2 \cdot \vec{n} = 0 \quad (7)$$

$$0 \cdot \vec{n} = 0 \quad (8)$$

$$16 \cdot \vec{n} = 0 \quad (9)$$

$$33 \cdot \vec{n} = 0 \quad (10)$$

Solution

Let

$$\vec{n} = a \quad (12)$$

$$b \quad (13)$$

$$c \quad (14)$$

$$a + 2b + 3c = 0, \quad (15)$$

$$3a + 3b + c = 0. \quad (16)$$

From these, we get

$$\vec{n} = t \begin{bmatrix} 7 \\ -8 \\ 3 \end{bmatrix} \quad (17)$$

$$-8 \quad (18)$$

$$3, \quad t \in \mathbb{R}, \quad t \neq 0 \quad (19)$$

Equation of plane is

$$\vec{n}^T \vec{x} = 1 \quad (20)$$

Solution

since point $\vec{p} = -132$ lies on the plain

$$\vec{n}^\top \vec{p} = 1 \quad (21)$$

Substituting,

$$7 - 8 \quad \quad \quad 3x \quad (22)$$

$$y \quad (23)$$

$$z = 7 - 8 \quad \quad \quad 3 - 1 \quad (24)$$

$$3 \quad (25)$$

$$2. \quad (26)$$

Solution

Substituting,

$$7 - 8 \qquad 3x \qquad (27)$$

$$y \qquad (28)$$

$$z = 7 - 8 \qquad 3 - 1 \qquad (29)$$

$$3 \qquad (30)$$

$$2. \qquad (31)$$

$$7 - 8 \qquad 3x \qquad (32)$$

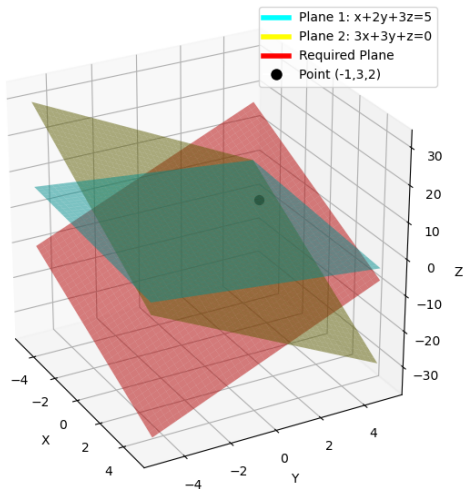
$$y \qquad (33)$$

$$z = -25 \qquad (34)$$

$$\frac{-1}{25} 7 - 8 \qquad 3x \qquad (35)$$

$$y \qquad (36)$$

Intersection of Planes and Required Perpendicular Plane



```
#ifndef PLANE_H
#define PLANE_H

#include <stdio.h>

// Function to compute cross product of two vectors
void cross_product(double a1, double b1, double c1,
                  double a2, double b2, double c2,
                  double *rx, double *ry, double *rz) {
    *rx = b1*c2 - c1*b2;
    *ry = c1*a2 - a1*c2;
    *rz = a1*b2 - b1*a2;
}
```


C Code

```
// Function to compute dot product of two vectors
double dot_product(double a1, double b1, double c1,
                   double a2, double b2, double c2) {
    return a1*a2 + b1*b2 + c1*c2;
}

// Function to print plane equation given normal and point
void plane_equation(double nx, double ny, double nz,
                   double x0, double y0, double z0) {
    double rhs = dot_product(nx, ny, nz, x0, y0, z0);
    printf("\nEquation of required plane: %.2lf*x + %.2lf*y + %.2lf*z = %.2lf\n",
           nx, ny, nz, rhs);
}

#endif
```

```
#include "solution.h"

int main() {
    double x0, y0, z0;
    printf("Enter point (x0 y0 z0): ");
    scanf("%lf %lf %lf", &x0, &y0, &z0);
    double a1, b1, c1, d1;
    printf("Enter coefficients of plane1 (a1 b1 c1 d1): ");
    scanf("%lf %lf %lf %lf", &a1, &b1, &c1, &d1);
    double a2, b2, c2, d2;
    printf("Enter coefficients of plane2 (a2 b2 c2 d2): ");
    scanf("%lf %lf %lf %lf", &a2, &b2, &c2, &d2);

    // Normal vectors of plane1 and plane2
    double n1x = a1, n1y = b1, n1z = c1;
    double n2x = a2, n2y = b2, n2z = c2;
```

```
// Compute required normal (cross product)
double nx, ny, nz;
cross_product(n1x, n1y, n1z, n2x, n2y, n2z, &nx, &ny, &nz)

// Print required plane
plane_equation(nx, ny, nz, x0, y0, z0);

return 0;

}
```

Python Code

```
import numpy as np

def cross_product(v1, v2):
    """Return cross product of two vectors"""
    return np.cross(v1, v2)

def dot_product(v1, v2):
    """Return dot product of two vectors"""
    return np.dot(v1, v2)

def plane_equation(normal, point):
    """Given normal vector and point, return plane equation coefficients"""
    d = dot_product(normal, point)
    return (*normal, d)
```

Python Code

```
def main():  
    # Input point  
    x0, y0, z0 = map(float, input("Enter point (x0 y0 z0): ").split())  
  
    # Input first plane coefficients  
    a1, b1, c1, d1 = map(float, input("Enter coefficients of plane 1: ").split())  
  
    # Input second plane coefficients  
    a2, b2, c2, d2 = map(float, input("Enter coefficients of plane 2: ").split())  
  
    # Normals of plane1 and plane2  
    n1 = np.array([a1, b1, c1])  
    n2 = np.array([a2, b2, c2])  
  
    # Required normal vector is cross product  
    n_required = cross_product(n1, n2)
```

```
# Given point
p = np.array([x0, y0, z0])

# Plane equation:  $n \cdot x = d$ 
a, b, c, d = plane_equation(n_required, p)

print(f"\nRequired plane equation:")
print(f"{a:.2f}x + {b:.2f}y + {c:.2f}z = {d:.2f}")

if __name__ == "__main__":
    main()
```

Python + C Code

```
import ctypes
import numpy as np

plane_lib = ctypes.CDLL("./solution.so")

plane_lib.cross_product.argtypes = [
    ctypes.c_double, ctypes.c_double, ctypes.c_double,
    ctypes.c_double, ctypes.c_double, ctypes.c_double,
    ctypes.POINTER(ctypes.c_double), ctypes.POINTER(ctypes.c_double)
]
plane_lib.cross_product.restype = None

plane_lib.dot_product.argtypes = [
    ctypes.c_double, ctypes.c_double, ctypes.c_double,
    ctypes.c_double, ctypes.c_double, ctypes.c_double
]
```

```
plane_lib.dot_product.restype = ctypes.c_double
```

```
plane_lib.plane_equation.argtypes = [  
    ctypes.c_double, ctypes.c_double, ctypes.c_double,  
    ctypes.c_double, ctypes.c_double, ctypes.c_double  
]
```

```
plane_lib.plane_equation.restype = None
```

```
def main():
```

```
    x0, y0, z0 = map(float, input("Enter point (x0 y0 z0): ").split())
```

```
    a1, b1, c1, d1 = map(float, input("Enter coefficients of p1: ").split())
```

```
    a2, b2, c2, d2 = map(float, input("Enter coefficients of p2: ").split())
```


Python + C Code

```
n1x, n1y, n1z = a1, b1, c1
n2x, n2y, n2z = a2, b2, c2
```

```
nx = ctypes.c_double()
ny = ctypes.c_double()
nz = ctypes.c_double()
plane_lib.cross_product(n1x, n1y, n1z, n2x, n2y, n2z,
    ctypes.byref(nx), ctypes.byref(ny), ctypes.byref(nz))
```

```
rhs = plane_lib.dot_product(nx.value, ny.value,
    nz.value, x0, y0, z0)
```

```
print(f"\nRequired plane equation:")
print(f"{nx.value:.2f}x + {ny.value:.2f}y +
    {nz.value:.2f}z = {rhs:.2f}")
```